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Paleomagnetism of the Sassamansville diabase, Newark Basin, southeastern Pennsylvania: Support for Middle Jurassic high-latitude paleopoles for North America

KENNETH P. KODAMA
ALLISON MOWERY
Department of Earth and Environmental Sciences, Lehigh University, Bethlehem, Pennsylvania 18015-3188

ABSTRACT

A paleomagnetic study of the Sassamansville diabase, which intrudes the Passaic Formation red beds of the Newark Basin in southeastern Pennsylvania, was conducted to constrain the age of the diabase's magnetization. Nine sites were collected from the diabase around the Sassamansville syncline. A tenth site was collected at an associated dike 12 km to the north. Eight to ten individually oriented cores were drilled at each site. Alternating field (af) and thermal demagnetization isolated a stable, internally consistent characteristic remanence at eight sites. Using the bedding tilt of the nearest baked or unbaked Passaic Formation sedimentary rocks for a fold test, the site means showed best clustering at 65% unfolding. This magnetization is a statistically significant (at 93% confidence) synfolding magnetization. The diabase's magnetization fails a baked contact test conducted at one of the sites, indicating that the magnetization is secondary and not a primary thermal remanent magnetization (TRM). Isothermal remanent magnetization (IRM) and partial anhysteretic remanent magnetization (pARM) acquisition experiments and af demagnetization results indicate that the magnetization is carried by high-coercivity magnetite. Anisotropy of magnetic susceptibility (AMS) and anhysteretic remanence (AAR) results indicate that this magnetite has a different fabric from the northeast-southwest horizontally lineated fabric carried by low-coercivity magnetite grains that do not carry the remanence. The lineated fabric may have been caused by flow of the diabase from the northeast to the southwest during emplacement. When the rock magnetic results are considered in light of previous petrographic observations (Davidson and Wyllie, 1968) of these rocks, the rock magnetics suggest that the magnetization is carried by secondary magnetite, which has grown as rims on primary magnetite. The secondary magnetite may have formed during a hydrothermal event proposed by Sutter (1988) based on geochronologic evidence and dated at 175 Ma. If the diabase magnetization is considered to be coeval to a widespread remagnetization (B component) of the Newark Basin sedimentary rocks (Witte and Kent, 1991), the diabase magnetization indicates 15° of counterclockwise rotation for the Sassamansville syncline. This is consistent with either left-lateral motion along the border fault or along the intrabasinal Chalfont Fault. The presence of counterclockwise block rotations along the border fault could weaken previously reported evidence that the Newark Basin (B component) remagnetization occurred after the Newark Basin strata had acquired most of their northwesterly tilt (Witte and Kent, 1991). Correction for the tilt would move the Newark (B component) remagnetization paleopole to lower paleolatitudes but necessitate significant post–175 Ma tectonic activity in the Newark Basin. This would contradict the long sedimentary record of rifting and border fault activity for the preceding 50 m.y. A partial tilt correction may be the best geologic and paleomagnetic interpretation because it would minimize the amount of post–175 Ma tectonic activity required while bringing the Newark (B component) remagnetization paleopole into paleolatitudinal agreement with a paleopole derived from the rotation of European, South American, and African data into North American coordinates (Van der Voo, 1992).

INTRODUCTION

Recent paleomagnetic studies of Mesozoic-age rocks from North America have led to two different interpretations about the position of the continent's paleomagnetic pole in the Middle Jurassic. May and Butler (1986) built on Gordon and other's (1984) approach of selecting only the most reliable paleopoles to construct a continent's apparent polar wander (apw) path and provided evidence for low-latitude (60°–70°) paleopoles for North America during the Middle Jurassic. This bolstered evidence for the paleomagnetic Euler pole (PEP) method of analyzing apparent polar wander paths (Gordon and others, 1984). In this approach it is assumed that plates rotate around one Euler pole with respect to the Earth's spin axis for geologically significant (tens to hundreds of million years) time periods. Apparent polar wander paths should therefore have a structure consisting of long tracks that are small circles around the Euler pole separated by sharp cusps caused when plate driving mechanisms changed and a new Euler pole was being established. An entirely different approach to apw path construction is to calculate a "running mean" of all the paleopoles available within a sliding time window (Irving and Irving, 1982). This approach will tend to smooth any sharp changes in apparent polar wander but will include a larger data set less affected by errors in any one paleopole. Irving and Irving's (1982) "running mean" apw path for North America indicated high-latitude (as much as 75°) paleopoles for the Middle Jurassic, primarily because of data from the White Mountain Magma Series of New England (Opdyke and Wensink, 1966). This study had been rejected by May and Butler because thermal demagnetization had not been employed, the paleopole was indistinguishable from geographic north, and the virtual geomagnetic pole (VGP) distribution was highly elongate, thus indicating the strong possibility of an unremoved present-day overprint. Van Fossen and Kent (1990) restudied the White Mountain igneous rocks (Moat Volcanics) and confirmed a high-latitude paleopole for Middle Jurassic North America; however, the magnetization they isolated failed a local tilt test, indicating it was a secondary magnetization, making the result controversial (Butler and others, 1992; Van Fossen and Kent, 1992). Witte and Kent

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(1991) have also isolated a secondary magnetization from Newark Basin sedimentary rocks that agrees with the Moat Volcanics paleopole from the White Mountains. They propose that the paleopole based on this secondary magnetization is 175–180 Ma by assuming that a hydrothermal event in the Newark Basin, postulated by Sutter (1988) from $^{39}Ar/^{40}Ar$ work on the Mesozoic intrusive rocks of the basin, caused a partial chemical remagnetization of the sedimentary rocks. Van der Voo (1992) has recently added to the discussion by rotating Middle Jurassic age paleopoles from South America, Europe, and Africa into the North American reference frame and suggesting that the North American paleopole lies somewhere between the high-latitude paleopoles of Kent and coworkers and the lower latitude paleopoles of May and Butler.

A better understanding of North America's Middle Jurassic apparent polar wander path is important because a high-latitude paleopole for the Middle Jurassic would essentially cut the long Jurassic apw "track" into smaller segments, thus bringing into question the fundamental assumptions and usefulness of PEP analysis. This would modify our understanding of plate dynamics.

The goal of this study is to re-examine the paleomagnetism and, in particular, the age of magnetization for the Newark intrusive rocks that contribute to the PEP apw path constructed by May and Butler (1986) and Gordon and others (1984). The Sassamansville diabase, which intrudes the sediments folded in the Sassamansville syncline in southeastern Pennsylvania, is part of the series of Mesozoic intrusive rocks that was used by Smith and Noltimier (1979, Appendix 1, Quakertown intrusive) (Beck, 1972) to determine their Newark Igneous Trend 2 (N2) paleopole. This paleopole was calculated from igneous rocks corrected for the bedding tilt of nearby sedimentary rocks. As Van Fossen and Kent (1990) pointed out, if the N2 magnetization is secondary and a paleopole is calculated from rocks not corrected for local tilt, the N2 paleopole is in agreement with both the Moat Volcanics and Newark secondary magnetization (B) paleopoles.

In this study, a fold test (Graham, 1949) will be used to constrain the age of magnetization of the Sassamansville diabase with respect to folding and thus determine the correct coordinate system from which to calculate the N2 paleopole. Previous paleomagnetic work (Kodama and others, 1993a) on the Passaic Formation red clastic sedimentary rocks folded in the Sassamansville syncline has isolated a prefolding, high unblocking temperature magnetization (C) and a synfolding, intermediate unblocking temperature magnetization (B) similar in character to the secondary magnetization isolated by Witte and Kent (1991) in other parts of the Newark Basin. The paleopoles calculated from these two magnetizations (B and C) are both rotated about 20° counterclockwise from coeval age paleopoles.

GEOLOGY

The Sassamansville diabase is part of the northern suite of eastern North American high TiO_2 quartz normative tholeiite dikes, sills, and flows (deBoer and others, 1988) which outcrop in the region's Mesozoic basins. Based on field relationships, bolstered by geophysical data (Ratcliffe, 1988; Kodama, 1983), that suggest a connection between the intrusive rocks and lava flows in the northern part of the Newark Basin and the cyclostratigraphy of sediments stratigraphically confining the flows, Olsen (1989) suggested a very short time period for the eruption of these igneous rocks (550,000 yr), despite the wide spread in radiometric ages reported for them (Sutter and Smith, 1979; McHone and Butler, 1984; deBoer and others, 1988). The best radiometric ages come from mineral separates (hornblende) that suggest crystallization at 201 Ma (Sutter, 1988). Sutter also provides evidence for a widespread low-temperature thermal event at about 175 Ma based on feldspars separated from granophyres and pegmatoids associated with the sills in both the Newark and Culpeper Basins. Sutter interprets this low-temperature event to be hydrothermal in origin, assuming that the diabase sheets cooled quickly from their initial emplacement and were not deeply buried subsequent to their eruption.

The Sassamansville diabase intrudes the Norian-age Passaic Formation at the Sassamansville syncline. Schlische (1989, 1992) termed the syncline a transverse fold, which was formed by differential amounts of slip along the Mesozoic basin border fault (Schlische, 1989, 1992). The pinching out of sedimentary units around the fold suggests that at least some of the folding was contemporaneous with deposition. The diabase appears to be conformal to sedimentary bedding in the syncline and cuts across

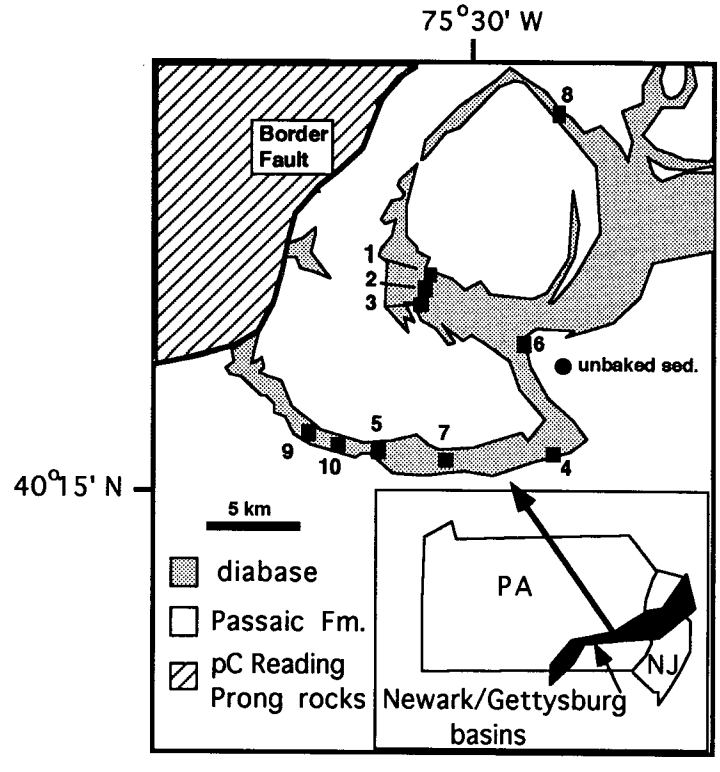


Figure 1. Geologic map showing locations of paleomagnetic sampling sites for the Sassamansville diabase. Solid dot shows location of sampling site of unbaked Passaic Formation sedimentary rocks for the baked contact test. Both baked sediments and diabase were sampled at site 6.

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TABLE 1. SITE LOCATIONS AND NEAREST SEDIMENTARY BEDDING ORIENTATIONS

Site	Latitude (°N)	Longitude (°E)	Bedding dip (°)	Bedding dip direction (°)
1	40.3517	284.4589	7	020
2	40.3464	284.4561	0	0
3	40.3435	284.4550	0	0
4	40.2679	284.5329	24	350
5	40.2671	284.4238	..	..
6	40.3560	284.5171	20	242
7	40.3514	284.4865	27	005
8	40.4200	284.5270	15	350
9	40.2700	284.3850	20	356
10	40.2700	284.4000	..	..

Note: sites 5 and 10 were not used for fold test; no bedding orientation was measured.

bedding in the adjacent anticline to the north (Fig. 1). This outcrop pattern would suggest that the diabase was intruded either during or after folding had taken place (Schlische, 1989). The outcrop pattern of the fold also indicates that the syncline plunges to the northwest. Based on the attitude of the hornfels and sedimentary beds measured near the diabase sampling sites, the fold axis trends 286° and plunges 5° to the northwest.

METHODS

We sampled the diabase at nine sites around the syncline and one site at a ring dike ~12 km to the north (Fig. 1, Table 1). We collected eight to ten individually oriented 25-mm-diameter cores at each site with a gasoline-powered diamond coring drill. The samples were oriented with a sun compass. This orientation was checked with a magnetic compass. The azimuths obtained by these two methods were typically only different by several degrees, but the difference could be as large as 10°. The samples were alternating field (af) demagnetized in a Schonstedt GSD-5 tumbling af demagnetizer in ten steps from 10 mT to 100 mT. To check the af demagnetization results, one sample from each site was completely thermally demagnetized in eight steps up to a temperature of 525 °C. All remanence measurements were made with either a Molspin spinner magnetometer or a CTF cryogenic magnetometer. Principal component analysis was used to calculate the characteristic remanence isolated from each sample (Kirschvink, 1980). Fisher (1953) statistics were used to determine the site mean directions. The fold test (Graham, 1949) was conducted by unfolding the site means around local strike in 5% steps of partial unfolding. The bedding attitude of the closest outcrop of baked or unbaked sediment to each diabase site was used for the tilt correction. Since sites 2 and 3 sampled the diabase at the hinge of the small anticline just north of the syncline (Fig. 1), where the outcrop pattern suggests the diabase cuts across bedding during intrusion, it was assumed that these sites had not been tilted tectonically during folding. Consequently their magnetizations were not unfolded during the fold test. The statistical significance of the mean direction with the highest precision parameter, k_{max}, was calculated (McFadden and Jones, 1981).

Three rock magnetic experiments were conducted to characterize the magnetic mineral that carried the remanence. To determine the samples' bulk coercivity isothermal remanence (IRM), acquisition experiments were conducted in DC magnetic fields up to 0.8 T for four samples from four sites. To estimate the efficiency of the magnetization in the samples as a method of determining whether the samples carried a primary thermal remanent magnetization (TRM), four samples' saturation IRM (IRM[s]) was af demagnetized and plotted versus the equivalent af demagnetization steps of their natural remanent magnetization (NRM) (Fuller and others, 1988). Finally, to measure the fabric in the diabase's magnetic grains, the anisotropy of magnetic susceptibility (AMS) was measured with a Sapphire Instruments SI-2 susceptibility meter for eight samples from eight sites. In addition, the anisotropy of anhysteretic remanence (AAR; McCabe and others, 1985) was measured in a 30–90 mT af window for seven of these samples. The AAR of one sample was also measured in a 0–20 mT af window to compare with its AMS results.

A baked contact test was also conducted at site 6 to determine whether the diabase's magnetization was primary. Five individually oriented cores were drilled from a nearby (40 m) hornfels outcrop. In addition, four oriented cores were collected from the nearest

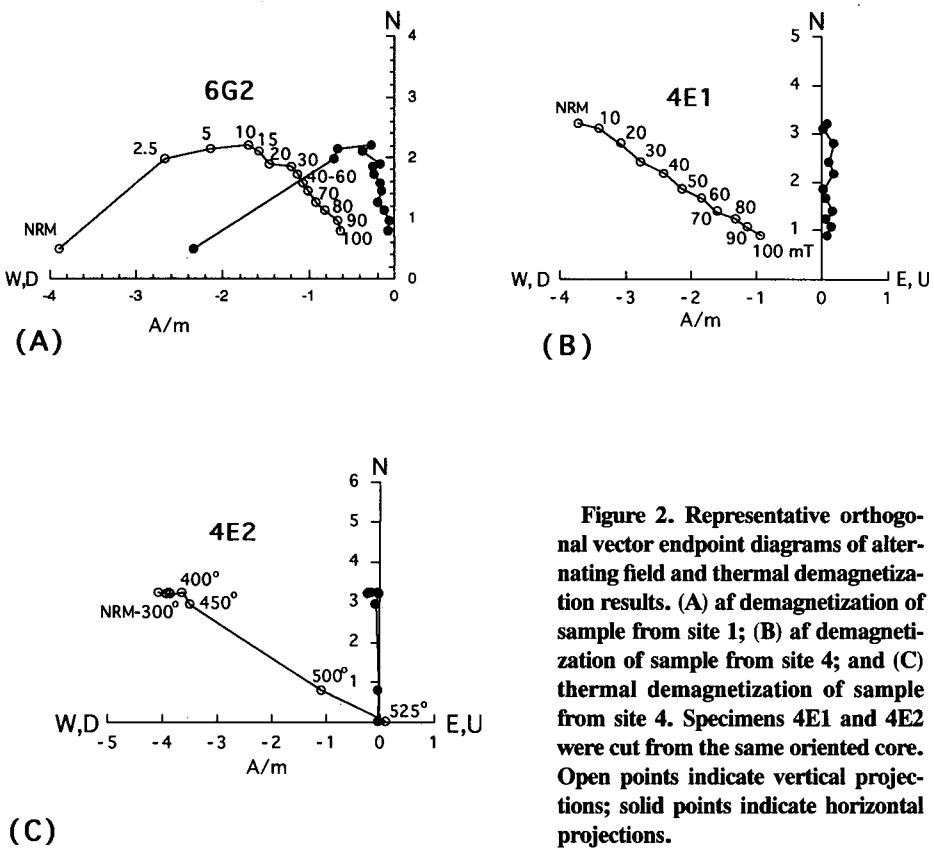


Figure 2. Representative orthogonal vector endpoint diagrams of alternating field and thermal demagnetization results. (A) af demagnetization of sample from site 1; (B) af demagnetization of sample from site 4; and (C) thermal demagnetization of sample from site 4. Specimens 4E1 and 4E2 were cut from the same oriented core. Open points indicate vertical projections; solid points indicate horizontal projections.

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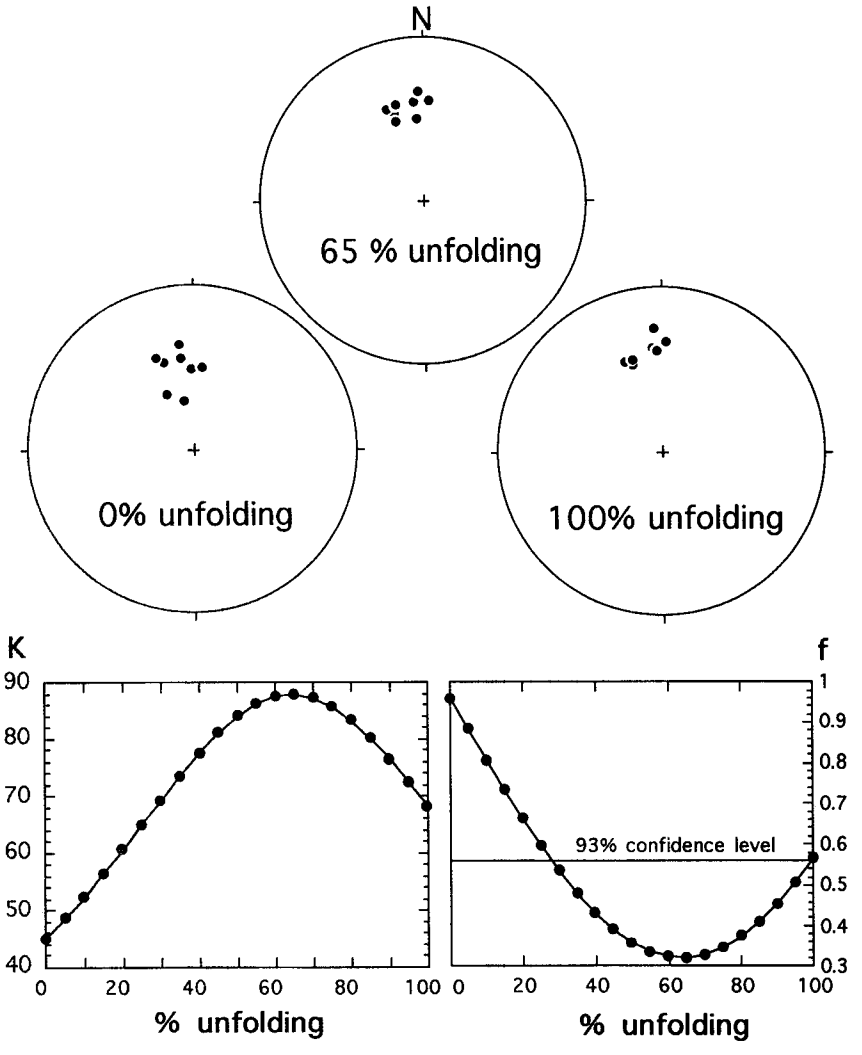


Figure 3. Partial, stepwise unfolding of diabase site means. Note that Fisher's (1953) precision parameter, k, shows maximum clustering at 65% unfolding and that McFadden and Jones's (1981) statistic, f, also indicates maximum clustering at 65% unfolding. The f statistic indicates that the mean of the site means at 65% unfolding (and all means between ~30% and 99% unfolding) are synfolding at the 93% confidence level. Stereonets are equal area; solid dots indicate lower hemisphere.

outcrop of unbaked Passaic Formation sediments, 2.5 km away. The zone of baked sediments is mapped (Longwill and Wood, 1965) to be ~2 km wide in this region.

RESULTS

Orthogonal vector endpoint diagrams (Zijderveld, 1967) of the alternating field demagnetization results indicate univectorial decay toward the origin after a low coercivity component has been removed below 10–50 mT. The high-coercivity, characteristic remanence in these rocks is normal polarity, with north to northwest declinations and moderately downward inclinations (Fig. 2). Thermal demagnetization of one sample per site also indicates univectorial decay to the origin up to a temperature of 525 °C (Fig. 2). The characteristic remanence isolated by thermal demagnetization is identical to that removed by af demagnetization. The results from sites 5 and 10, both from the southern limb of the syncline, were scattered and removed from further analysis. Site 10 was sampled from three different boulders whose tops were exposed at a new construction site, and it appears these boulders were not attached to bedrock. Site 5 had anomalous west-northwesterly directions scattered along a small circle, and the rocks, at the sampling site, were fractured. A fault is mapped in the surrounding sedimentary rocks (Longwill and Wood, 1965). Apparently it cuts through the diabase at this locality and may have caused the scatter.

When the remaining eight site means are partially unfolded around local strike in 5% steps, the maximum precision parameter, k_{max}, occurs at 65% unfolding (Fig. 3, Table 2). The mean direction with maximum k is distinguishable from the *in situ* case at 97.5% confidence level and the totally unfolded case at the 93% confidence level (McFadden and Jones, 1981) and is interpreted to be synfolding in age. Because of the small plunge of the fold axis (5°), the mean direction at 65% unfolding changes by <2° when the fold is unplunged and then unfolded. If the site means are unfolded around the fold axis, the maximum clustering of site means also occurs at 65% unfolding. If the northernmost site, 8, which is not on the syncline, is excluded from the fold test, k_{max} also occurs at 65% unfolding. However, the k_{max} magnetization can be distinguished statistically from the 100% unfolding case at only the 80% confidence level but can be distinguished from the *in situ* case at 99% confidence. Therefore, if site 8 is excluded from the fold test, the magnetization is either syn- or prefolding, but

TABLE 2. PALEOMAGNETIC SITE MEAN DIRECTIONS IN GEOGRAPHIC COORDINATES

Site	N/N_o	Inclination (°)	Declination (°)	k	α_{95} (°)
1	10/10	43.6	352.7	152.5	3.6
2	9/9	44.3	341.3	354.1	2.5
3	9/9	40.4	338.1	14.0	12.5
4	9/10	49.6	359.5	318.6	2.5
5	8/8	33.3	284.9	27.1	9.5
6	9/9	36.3	353.4	801.5	1.7
7	8/8	65.6	350.7	121.5	4.5
8	8/8	48.1	006.0	82.8	5.4
9	5/9	60.3	335.2	105.9	6.1
10	8/8	16.6	023.6	2.5	31.6

Note: sites 5 and 10 were not used for fold test; see text for discussion. N is the number of samples used to calculate the site mean direction; N_o is the number of samples collected at a site.

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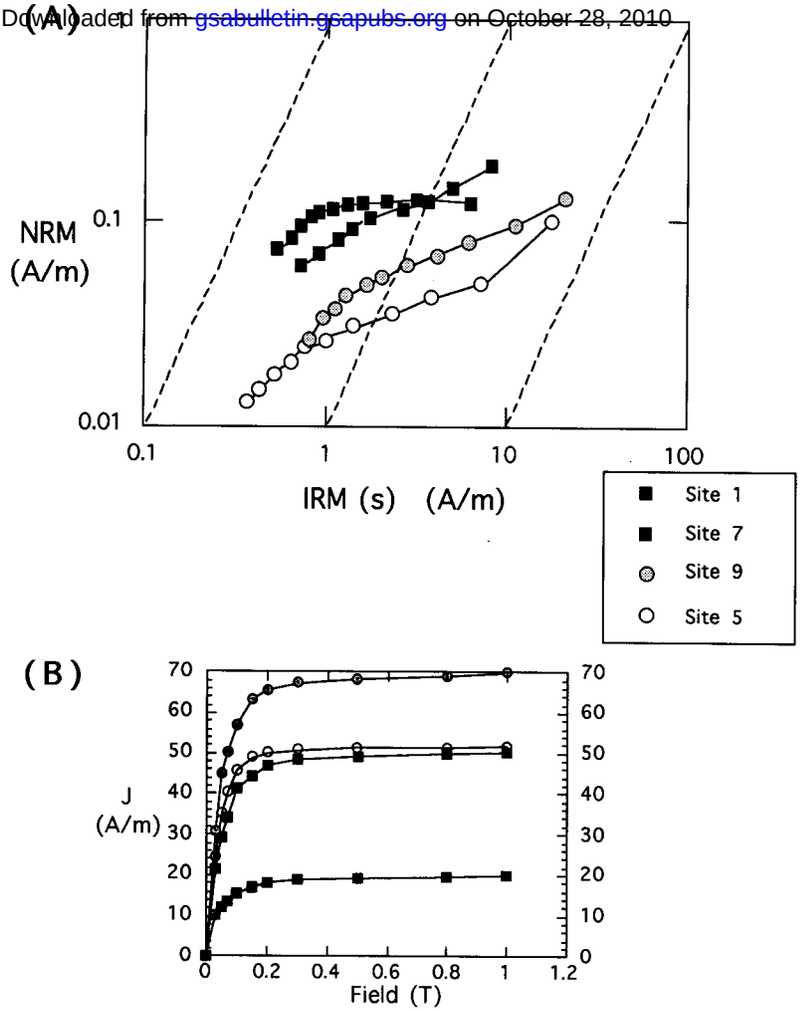


Figure 4. (A) Alternating field demagnetization of natural remanent magnetization (NRM) plotted as a function of af demagnetization of saturation isothermal remanence (IRM[s]) to measure the efficiency of the natural remanence of the samples (Fuller and others, 1988). This may be used to indicate a primary thermal remanence (TRM) in the samples. Each set of connected points represents the af demagnetization of one sample with af fields progressing from 0 in the upper right to 100 mT in the lower left of the plot. The dashed lines indicate constant ratios of NRM:IRM(s). The results indicate an efficient magnetization consistent with a primary TRM; however, the baked contact test results (see text and Fig. 6) suggest that magnetization is not primary. (B) Isothermal remanent magnetization (IRM) acquisition experiment results for samples from four sites. Note that the magnetization saturates by fields of 0.2 T, which suggests magnetite is the primary magnetic carrier for these rocks.

not postfolding in age. The paleomagnetic pole calculated from the synfolding magnetization for all eight sites occurs at 71.8°N, 136.5°E with A_{95} = 5.8°. When site 8 is excluded, the paleopole lies at 71.2°N, 142.0°E, A_{95} = 5.5°. Because a synfolding magnetization only constrains the relative position of the fold limbs at the time of remanence acquisition, these paleopoles are only accurate if no undetected coherent tilting of the Newark Basin occurred subsequent to remanence acquisition.

Rock magnetic results indicate that the magnetic carrier is probably low Ti titanomagnetite or magnetite. This is based on the IRM acquisition experiments in which saturation remanence is reached by 0.3 T (Fig. 4) and the thermal demagnetization in which the remanence is removed by 525 °C. When the af demagnetization of NRM is plotted versus the af demagnetization of IRM(s), an NRM: IRM(s) ratio of about 0.1 results, indicating a high efficiency of magnetization, which could be interpreted as a TRM (Fig. 4; Fuller and others, 1988). The AMS has an average anisotropy of 3.5% and shows a pronounced northeast-southwest, nearly horizontal lineation in stratigraphic coordinates. This fabric clusters best in tilt-corrected coordinates; however, the clustering is not statistically significant (Fig. 5, Table 3). The coercivity spectrum indicated by the partial anhysteretic remanent magnetization (pARM) experiments shows that most of the power lies in low coercivities below 20 mT (Fig. 5). However, based on the af demagnetization results, the remanence is carried by grains with higher coercivities. The magnetic fabric for these high-coercivity remanence-carrying grains is measured by AAR applied in a 30–90 mT window and shows a fabric with a nearly vertical lineation, which is much different from the AMS fabric (Fig. 5, Table 4). In order to check for a possible inverted fabric due to the presence of single domain (SD) grains (Potter and Stephenson, 1988; Rochette, 1988), the AAR of one sample from site 6 was also measured in a 0–20 mT window. This fabric is identical to the AMS fabric for this sample, suggesting that the AMS fabric is due to large, probably multidomain grains and not an inverted fabric due to SD grains, which probably carry the high-coercivity fabric. This indicates that the two different fabrics are caused by two different populations of magnetic grains.

In the baked contact test, thermal demagnetization of the hornfels samples shows univectorial decay toward the origin up to temperatures of 645 °C (Fig. 6). This is consistent with the hematite-bearing red beds of the Passaic Formation being thermally overprinted by the intrusion of the diabase. However, the characteristic magnetization isolated from the hornfels site (Inc. = 5.0°; Dec. = 13.7°; α_{95} = 5.0°) is statistically different from the site mean direction (site 6) of the nearby diabase (Fig. 6). The mean direction for the site collected from the nearest unbaked sediment (Inc. = −14.5°; Dec. = 140.6°; α_{95} = 8.5°) is reversed polarity and similar to the northwest-southeast dual polarity, shallow inclination characteristic magnetizations isolated from the Passaic Formation at the Sassamansville syncline (Kodama and others, 1993a). A separate paleomagnetic study of the sediments baked by the Sassamansville diabase (Kodama, 1993; Kodama and others, 1993b) indicates that the hornfels can have two different magnetic behaviors. Dark gray hornfels is strongly magnetized and contains hematite, identical to the magnetic characteristics of the baked contact test near diabase site 6. This magnetization is prefolding and

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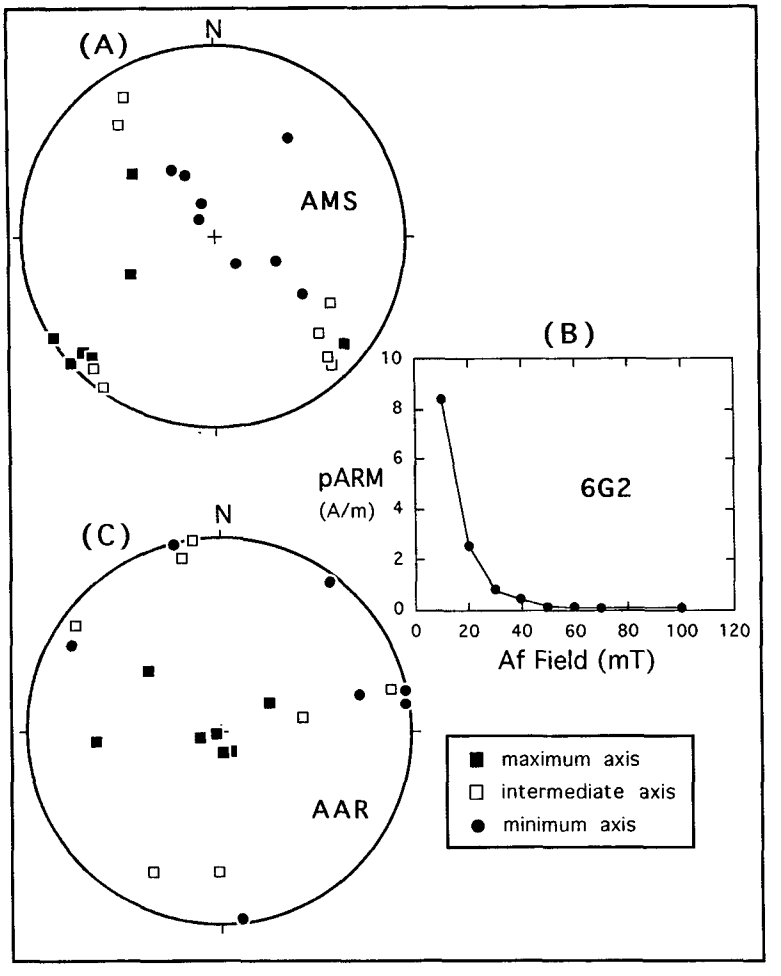


Figure 5. (A) Anisotropy of magnetic susceptibility (AMS) results for the Sassamansville diabase, indicating a strong subhorizontal, southwest-northeast lineation, probably indicating that the magma flowed to the southwest. (B) Acquisition of partial anhysteretic remanence (pARM) experimental results, indicating that most of the grains are large, low-coercivity (<20 mT) grains that do not carry the remanence (see Fig. 2 and text). These large grains probably dominate the AMS results. (C) Anisotropy of anhysteretic remanence (AAR) results for pARMs acquired between 30 and 90 mT indicate a magnetic fabric for the remanence-carrying grains that is different from the fabric carried by the large grains responsible for the AMS. All points are lower hemisphere projections on the equal-area stereonets.

significantly different from the mean diabase magnetization reported in this paper. These results bolster the baked contact test reported here and suggest that the diabase magnetization is not primary. Light gray beds in the hornfels are characterized by a weak magnetization, which is similar in direction to the diabase magnetization and is carried by magnetite. The light gray hornfels magnetization clusters best at partial unfolding, but this clustering is not statistically significant.

DISCUSSION

The failure of the baked contact test by the diabase's magnetization is particularly intriguing because the rock magnetic evidence could be consistent with a primary TRM. The IRM acquisition data indicating magnetite as the magnetic carrier is certainly expected for a primary TRM of a mafic intrusive rock. The high magnetic efficiencies implied by Fuller and others' (1988) test are also consistent with a primary TRM. However, the results of this test may be tainted by the presence of two populations of magnetite indicated by the magnetic fabric measurements. The large magnetite grains carrying what is apparently a prefolding northeast-southwest lineation are quite possibly primary and may give a relatively high efficiency result. These grains, based on the af demagnetization results, do not carry the stable remanence of the diabase. Their fabric probably resulted from the original flow of the diabasic magma as it was being emplaced either due to actual mechanical orientation of the magnetite grains during flow (for example, Khan, 1962; Elwood, 1978; Knight and Walker, 1988) or the imposition of fabric on the magnetite as it crystallized in a flow-oriented silicate template after flow had ceased (Hargraves and others, 1991). The most recent observational work correlating macroscopic flow lineations in mafic dikes with AMS fabric suggests that magnetic lineations are parallel to magma flow (Knight and Walker, 1988; Staudigel and others, 1992). If the large ring dike to the northeast of the Sassamansville syncline (Fig. 1) indicates the magma source area, this would suggest that the flow was from the northeast toward the southwest, subparallel to the border fault. The high-coercivity magnetite grains which do carry the stable remanence in the diabase have an entirely different magnetic fabric, suggesting that they are a different population of magnetic grains and are secondary. Detailed opaque petrographic studies of the diabasic rocks from southeastern Pennsylvania support this interpretation for the genesis of the magnetic grains in the Sassamansville diabase. Davidson and Wyllie (1968) report two forms of magnetite in the diabase and associated granophyres. One form contains exsolved ulvospinel and is closely associated with ilmenite. The second form is Ti poor and contains no microscopically visible Ti-bearing phases. It occurs as rims on the first form of magnetite or as separate crystals in the more granophyric rocks. It tends to be associated with skeletal ilmenite, which retains an original spinel outline. Davidson and Wyllie speculate that these observations point to an aqueous fluid developing within the magma, which dissolved iron from the earlier Ti-magnetite intergrowths and left skeletal crystals of ilmenite. They suggest that some or all of this iron was redeposited as a second generation of Ti-poor magnetite. Davidson and Wyllie attribute the aqueous fluids to magmatic differentiation processes; however, based on the paleomagnetic and rock magnetic evidence reported here and Sutter's (1988) geochronologic observations, we propose a different scenario.

Early in the folding process, or even before folding began, the diabase intruded into the red clastic sediments of the Passaic Formation. This is based on the prefolding hornfels magnetization, which is reported in more detail in a separate paper (Kodama and others, 1993b). The observation that the diabase is

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TABLE 3. DIABASE AMS RESULTS IN GEOGRAPHIC COORDINATES

Sample	Principal axis	Inclination (°)	Declination (°)	Eigenvalue (1/4p × 10^{-4}) SI
1d1	Min.	42.6	38.7	1.61
	Inter.	10.5	138.5	1.63
	Max.	45.4	239.4	1.65
2e1	Min.	61.9	112.5	5.61
	Inter.	25.3	320.1	5.74
	Max.	11.4	224.7	5.82
3c2	Min.	75.0	142.2	5.31
	Inter.	14.9	327.7	5.50
	Max.	1.4	237.3	5.67
4f2	Min.	75.9	292.7	2.20
	Inter.	13.7	127.3	2.22
	Max.	03.4	36.4	2.25
6g2	Min.	69.9	34.7	8.98
	Inter.	4.0	135.7	9.12
	Max.	19.7	227.1	9.30
7h	Min.	76.4	267.8	4.74
	Inter.	8.4	139.9	4.87
	Max.	10.6	48.4	5.00
8b2	Min.	33.0	131.7	12.08
	Inter.	8.2	36.4	12.22
	Max.	55.8	294.2	12.33
9d1	Min.	83.2	210.9	6.12
	Inter.	06.7	42.1	6.21
	Max.	1.3	312.0	6.32

conformal to sedimentary bedding in the syncline, but not in the adjacent anticline, suggests that some folding had occurred at the time of diabase intrusion (Schlische, 1989). The strong prefolding configuration of the hornfels magnetization, however, rules out large bedding dips at this time. The AMS results suggest that the primary magnetite (the first form described by Davidson and Wyllie [1968]) recorded the flow of the diabase during emplacement. Sutter's $^{39}Ar/^{40}Ar$ work on hornblende separates indicates that this occurred at about 201 Ma. Subsequently, the hydrothermal fluids proposed by Sutter (1988) flowed through the rocks at about 175 Ma. The evidence for a widespread hydrothermal event is based on argon closure ages for potassium feldspars separated from the granophyric rocks associated with diabase in the Newark and Culpeper Basins. The fluids associated with this hydrothermal event caused partial dissolution of Davidson and Wyllie's first form of magnetite and its associated ilmenite and deposited the second form of fine-grained, remanence-carrying, Ti-poor, secondary magnetite as rims on the primary magnetite. The secondary magnetite acquired a chemical remanent magnetization (CRM) during the folding of the Passaic formation and the diabase at the Sassamansville syncline. Based on the high coercivity AAR results, this magnetite has an entirely different magnetic fabric than the primary magnetite grains. The hydrothermal fluids also caused the partial remagnetization of the surrounding Passaic Formation red beds, and this component of magnetization has been observed as the Newark remagnetization (B component) by Witte and Kent (1991) and Kodama and others (1993a). The remagnetization of the diabase and red beds most likely occurred when North America was at high latitudes.

The paleomagnetic results from the Sassamansville diabase record one more tectonic event, which occurred subsequent to the flushing of remagnetizing hydrothermal fluids through the region. The paleopole from the mean diabase magnetization is displaced from the coeval age B component remagnetization paleopole reported for the Newark Basin (Witte and Kent, 1991) (Fig. 7). If the synfolding diabase paleopole based on all eight sites is compared to the Newark remagnetization (B component) paleopole, it suggests that the Sassamansville syncline has been rotated counterclockwise by 13.0° (±5.4°) and has not been shifted significantly in latitude (0.08° ± 4.9°) (Demarest, 1983). If the northernmost site, which is not located in the syncline, is removed from the data set, then a 15.1° (±5.2°) ccw rotation is suggested and again no latitudinal offset is indicated (−0.2° ± 4.7°). These results confirm the ~19° ± 6° counterclockwise rotations observed for the B and C magnetization components isolated from the Passaic Formation red beds at the Sassamansville syncline (Kodama and others, 1993a) and indicate that a 15° counterclockwise block rotation occurred at the Sassamansville syncline subsequent to the hydrothermal event at 175 Ma. A counterclockwise block rotation of this area, which lies along the border fault, is consistent with geologic models that postulate left-lateral shear along the border fault during formation of the Newark Basin (Lucas and others, 1988). It may also be consistent with left-lateral motion along the Chalfont Fault, a large intrabasinal fault, which terminates just to the northeast of the Sassamansville syncline. Schlische (1992) has provided geologic evidence for this type of motion along the Chalfont Fault.

The possibility of vertical axis block rotations for the transverse folds along the border fault necessitates reexamination of Witte and Kent's interpretation of a posttilting age for the Newark (B component) remagnetization. Their evidence for a posttilting age is primarily a result of the more northwesterly trend of the Jacksonwald fold axis when compared to the dip direction of the Newark Basin strata. As the homoclinally dipping Newark Basin strata and Jacksonwald fold axis are untilted, their B component magnetizations diverge. If a counterclockwise block rotation should be removed from the transverse folds along the

TABLE 4. DIABASE AAR RESULTS IN GEOGRAPHIC COORDINATES FOR pARM APPLIED IN A 30–90 mT COERCIVITY WINDOW

Sample	Principal axis	Inclination (°)	Declination (°)	Eigenvalue (A/m)
1e2	Min.	3.9	353.6	0.416
	Inter.	57.0	89.7	0.425
	Max.	32.7	261.0	0.455
2e1	Min.	2.0	81.8	0.467
	Inter.	0.8	351.8	0.490
	Max.	87.8	239.2	0.506
3c2	Min.	26.0	75.7	0.644
	Inter.	28.0	180.7	0.704
	Max.	50.0	310.0	0.731
4f2	Min.	26.9	293.1	0.571
	Inter.	0.3	203.0	0.631
	Max.	63.1	112.4	0.658
6g2	Min.	14.1	216.0	0.299
	Inter.	16.5	310.2	0.305
	Max.	80.7	151.8	0.339
7h	Min.	26.0	344.2	0.553
	Inter.	16.3	82.4	0.567
	Max.	58.7	201.1	0.590
9d1	Min.	3.5	78.6	0.350
	Inter.	29.0	346.7	0.359
	Max.	60.7	174.8	0.407

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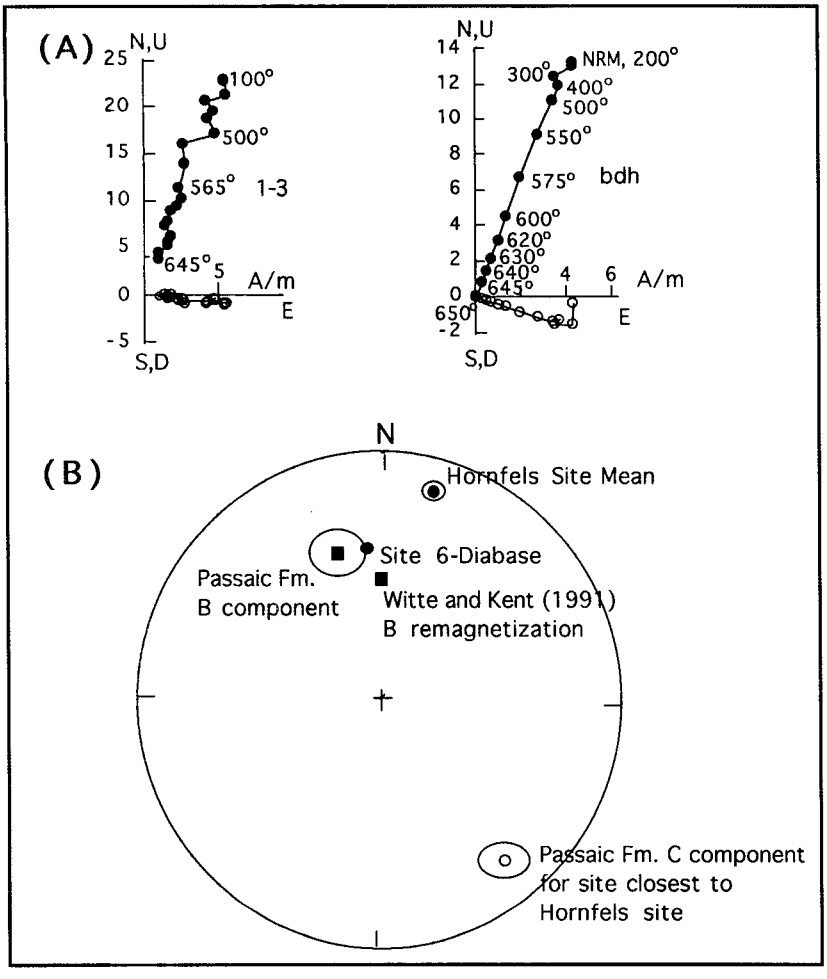


Figure 6. (A) Representative orthogonal vector endpoint projections of thermal demagnetization results for baked sediments (hornfels) near diabase site 6. Unblocking temperatures as high as 650 °C suggest that the hematite of the Passaic Formation red beds has been thermally reset by the intrusion of the diabase. (B) Directional results for the hornfels site, the nearby diabase site 6, and the nearest unbaked Passaic Formation sedimentary rocks on an equal-area stereonet. The hornfels site mean and diabase site mean are statistically different at the 99% confidence level. The diabase site mean is close to both the secondary, synfolding B component magnetization isolated in the nearby Passaic Formation sediments by Kodama and others (1993a) and to the B component remagnetization observed by Witte and Kent (1991) throughout the entire Newark basin. These results suggest that the hornfels magnetization is a TRM acquired during intrusion of the diabase, but the diabase magnetization has been reset by the same event that imparted the B component magnetization to the sedimentary rocks of the Newark basin. The hornfels magnetization at this location was apparently not reset. Open circles are upper hemisphere projections. Solid circles are lower hemisphere projections.

border fault, the fold axis trends will be closer to the dip direction of the Newark Basin strata. If a tilt correction is then applied, the B component magnetization carried by the Jacksonwald fold will neither diverge from nor converge toward the B component magnetization of the basin strata. Hence, the age of the B component would not be well constrained.

Correcting the Newark (B component) remagnetization paleopole for 10°–15° of northwesterly tilt typically found in the basin would move the paleopole closer to May and Butler's (1986) apparent polar wander path for North America (to 66°N, 120.9°E for the maximum of 15° of dip). Although this scenario could remove evidence for Middle Jurassic high-latitude paleopoles for North America, it would also necessitate significant post–175 Ma tectonic tilting in the Newark Basin. This is unlikely because the stratigraphy of the basin indicates that rifting, and hence activity on the border fault, was relatively continuous from its initiation at 230 Ma (Schlische, 1992). Perhaps the most likely interpretation is that some, but not all, of the 15° homoclinal dip of the Newark strata could have occurred after the Newark Basin remagnetization event. In this case the late Middle Jurassic paleopole derived from the rotation of European, South American, and African data into North American coordinates (Van der Voo, 1992) may be the most reasonable interpretation for the location of Middle Jurassic, North American paleopoles.

CONCLUSIONS

Based on a fold test and the failure of a baked contact test, the Sassamansville diabase, which intrudes the Norian-age Passaic Formation in the southeastern Newark Basin, has a secondary, synfolding-age magnetization. The synfolding nature of the diabase magnetization indicates that Smith and Noltimier's Newark Trend 2 (N2) paleopole should also be calculated from synfolding coordinates. If this magnetization is inferred to be coeval to the Newark remagnetization paleopole of Witte and Kent (1991), it suggests a 15° counterclockwise rotation of the Sassamansville syncline with respect to the main part of the Newark Basin. This sense of rotation is consistent with either a left-lateral offset along the border fault during basin formation or left-lateral offset of the nearby Chalfont Fault. Although the presence of counterclockwise block rotations along the border fault could reopen the possibility that the 175 Ma Newark (B component) remagnetization should be corrected for the northwesterly tilt of the Newark Basin strata, this would require significant tectonic tilting very late in the basin's history and contradict the long sedimentary record of earlier rifting and tilting. Therefore, the Newark remagnetization supports Middle Jurassic moderately high-latitude paleopoles for North America.

The secondary magnetization of the diabase is carried by high-coercivity magnetite, which has a different magnetic fabric from the low-coercivity magnetic grains whose lineated fabric may indicate the original flow of the diabase from the southwest to the northeast during emplacement. When the rock magnetic results are interpreted in light of Davidson and Wyllie's (1968) petrographic examination of southeastern Pennsylvania diabase and granophyre, they suggest that the

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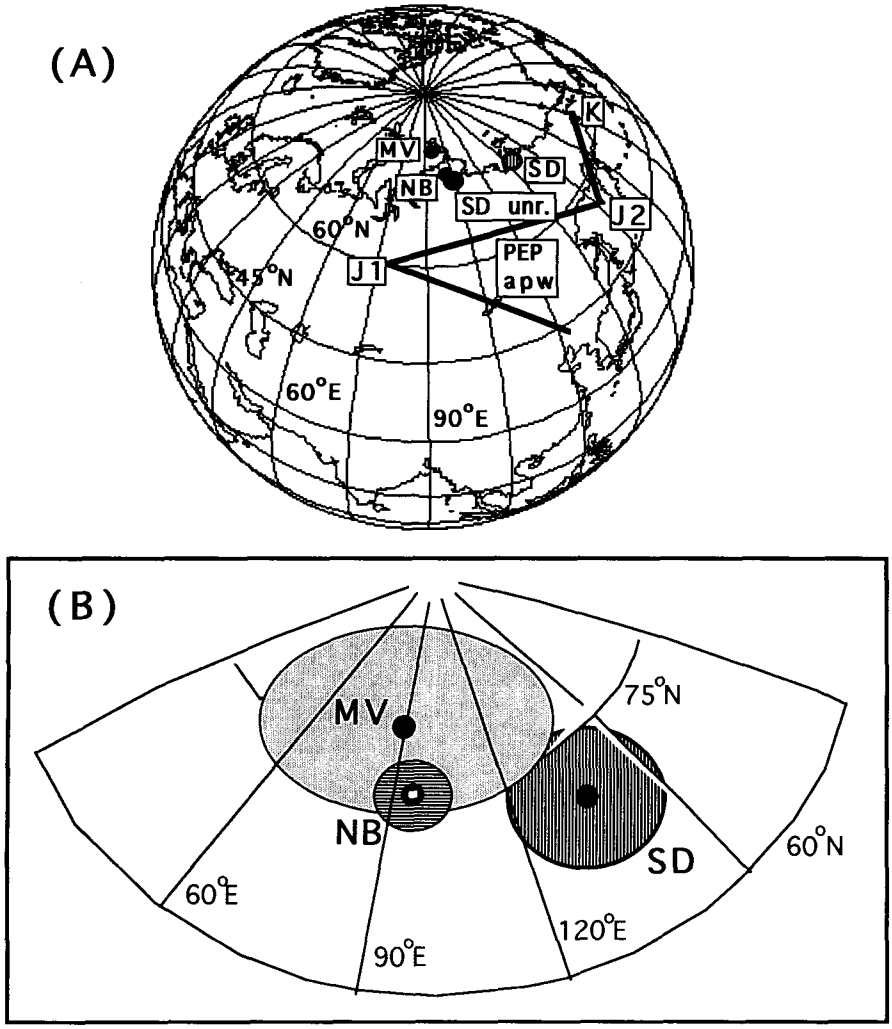


Figure 7. Paleomagnetic poles for the Sassamansville diabase. (A) The location of the Sassamansville diabase (SD) paleopole compared to the Moat Volcanics (MV) pole of Van Fossen and Kent (1990) and the Newark B remagnetization (NB) pole of Witte and Kent (1991). When a 15° counterclockwise rotation around a vertical axis at the site is removed from the Sassamansville data, the Sassamansville unrotated pole (SD unr.) lies directly on the NB pole (see B for more detailed plot). Solid lines represent a simplified version of the paleomagnetic Euler pole (PEP) apparent polar wander (apw) path proposed by Gordon and others (1984) and May and Butler (1986). J1 and J2 are Early and Late Jurassic, respectively, cusps proposed for the pole path that indicate a significant change in plate motion. The SD and SD unr. poles are not in agreement with the PEP apw path but are consistent with high-latitude poles (MV and NB) for the Middle Jurassic. K = Cretaceous. (B) Plot indicating 95% confidence limits for the poles shown in A. The open square in the center of the NB pole is the location of the unrotated SD pole.

magnetization is carried by a low-Ti secondary form of magnetite that grows as rims on a primary form of magnetite that contains exsolved ulvospinel. If the secondary magnetite acquired a CRM during its growth from fluids during a hydrothermal event suggested by Sutter's (1988) $^{39}Ar/^{40}Ar$ geochronologic work in the Newark Basin the best estimate for the age of the magnetization is 175 Ma.

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